

Working Out Specific Numericals

Monday, 20 July 2026

10:36 PM

$$x_0 = -26\ 072\ 138.45\ \text{Km}$$

$$y_0 = 144\ 774\ 673.82\ \text{Km}$$

$$v_{x_0} = -29.79\ \text{Km/sec}$$

$$v_{y_0} = -5.40\ \text{Km/sec}$$

$$\Delta t = 3600\ \text{s}$$

$$Gm_{\text{sun}} = 1.327 \times 10^{11}\ \text{Km}^3/\text{s}^2$$

$$r = \sqrt{x_0^2 + y_0^2}$$

$$= \sqrt{(-26\ 072\ 138.45)^2 + (144\ 774\ 673.82)^2}$$

$$\approx \sqrt{6.7976 \times 10^{14} + 2.0960 \times 10^{16}}$$

$$\approx \sqrt{2.1640 \times 10^{16}}$$

$$\approx 147\ 105\ 000\ \text{Km}$$

Sanity check: The distance between the earth is ≈ 150 million Km, and my calculated value is a very close approximation.

$$\frac{Gm_{\text{sun}}}{r^2} = \frac{1.37 \times 10^{11}}{147\ 105\ 000^2} = \frac{1.37 \times 10^{11}}{2.1640 \times 10^{16}}$$

$$\approx 6.132 \times 10^{-6}\ \text{Km/s}^2$$

$$\frac{x_0}{r} = -\frac{26\ 072\ 138.45}{147\ 105\ 000} \approx -0.17724$$

$$\frac{y_0}{r} = \frac{144\ 774\ 673.82}{147\ 105\ 000} \approx 0.98416$$

$$a_x = -\left(\frac{Gm_{\text{sun}}}{r^2}\right)\left(\frac{x_0}{r}\right)$$

$$= -(6.132 \times 10^{-6})(-0.17724)$$

$$\approx 1.087 \times 10^{-6}\ \text{Km/s}^2$$

$$a_y = -\left(\frac{Gm_{\text{sun}}}{r^2}\right)\left(\frac{y_0}{r}\right)$$

$$= -(6.132 \times 10^{-6})(0.98416)$$

$$\approx -6.035 \times 10^{-6}\ \text{Km/sec}^2$$